

Dhairya Joshi

B.Tech., Indian Institute of Technology (IIT), Jodhpur (Minor in AI)
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EDUCATION

B.Tech – Indian Institute of Technology Jodhpur – CGPA: 8.43	2024 – 2028
Senior Secondary – Percentage: 92.2%	2024
Secondary – Percentage: 96.4%	2022

EXPERIENCE

Undergraduate Research Assistant – FinSense AI	GitHub	Dec. 2025 – Jan. 2026
<i>Supervisor: Prof. G. Venkat Ram Reddy, SME, IIT Jodhpur</i>		
- Engineered 80+ leakage-safe features from market data, news sentiment, and Google Trends signals across 20+ stocks, lowering next-day return prediction error by 20% via a weighted ensemble of Random Forest, Gradient Boosting, and XGBoost models. - Deployed a FastAPI backend exposing REST APIs to a live frontend with real-time progress tracking, serving predictions across the full tracked ticker universe alongside an event-driven backtesting engine (fixed/trailing/ATR stop-loss) evaluated via Sharpe ratio, max drawdown, and win-rate. - Diagnosed and resolved production JSON serialization failures from NaN/Inf propagation in external financial APIs, eliminating recurring backend crashes.		

PROJECTS

CheckerPlane AI – Low-Latency LLM Guardrail Proxy	GitHub	Aug. 2026
<i>Rust, ONNX Runtime, C++, DeBERTa, BERT, Tokio, Async Concurrency</i>		
- Built a multi-tier AI safety proxy in pure Rust with ONNX Runtime, replacing sequential Python guardrail chains with parallel async gate execution, cutting median guardrail latency from ~ 550ms to ~67ms (~8x reduction). - Engineered a 4-gate parallel input rail (toxicity, PII/Aadhaar detection via BERT-based NER + regex, coherence scoring, prompt-injection defence) with early-exit cancellation, and a DeBERTa cross-encoder NLI hallucination gate scoring output grounding pre-delivery. - Achieved ~88% average gate accuracy across five safety gates with zero GPU dependency using quantised ONNX models; added a Tier-0 heuristic screen (<0.1ms) with configurable per-gate thresholds and fail-open policy.		
Delta Hedging & Adversarial Stock Path Generation Simulator	GitHub	July 2026 – Aug. 2026
<i>Python, TensorFlow, NumPy, Pandas, Monte Carlo Simulation, Black-Scholes Model</i>		
- Built a quantitative finance simulator benchmarking Black-Scholes delta hedging against deep learning-based hedging strategies across 10,000+ synthetic market scenarios generated through Monte Carlo simulation. - Architected and trained a GRU-based deep hedging model using a custom Conditional Value-at-Risk (CVaR) loss function, reducing CVaR_{5%} tail risk from Rs. 500 (Black-Scholes baseline) to Rs. 290, a 42% improvement. - Designed an adversarial training framework where a recurrent generator learns worst-case market conditions, further cutting CVaR_{5%} to Rs. 100 – an 80% drop versus the Black-Scholes baseline. - Processed 8,500+ minute-level NIFTY options data points, computing implied volatility and Black-Scholes Greeks for model calibration and evaluation.		
LLM Response Preference Classification – Kaggle Competition	GitHub	Sept. 2026
<i>Python, PyTorch, HuggingFace Transformers, Llama 3, Gemma 2, xFormers, Multi-GPU Inference</i>		
- Fine-tuned Llama-3 and Gemma-2 sequence-classification models on ~57,400 datapoints to predict human preference between pairs of LLM chatbot responses, framing the task as a 3-class (win A / win B / tie) classification problem. - Engineered a pipelined multi-GPU inference system splitting each Llama-3 model across two GPUs, using variable-length attention (xFormers block-diagonal causal masking) and sharded max-token batching to eliminate padding waste and maximize throughput. - Combined Llama-3 and Gemma-2 predictions with response-order swapping as test-time augmentation via weighted ensembling, achieving a 0.83040 log-loss and, competing solo, placing 8th of 224 teams (top 4%).		

TECHNICAL SKILLS

Languages: Python, C++, SQL, Rust
Machine Learning: Scikit-Learn, Random Forest, XGBoost, Gradient Boosting, Regression, Classification, Feature Engineering, Time-Series Forecasting
Deep Learning: PyTorch, TensorFlow, Keras, ANN, CNN, RNN, LSTM, GRU, GANs, Transformer Architectures (Llama, Gemma, BERT, DeBERTa)
Generative AI / LLMs: LLM Fine-Tuning, RAG, LangChain, Vector Databases, Prompt Engineering, Agentic AI Systems, LLM Guardrails & Safety Pipelines
Backend & Systems: FastAPI, REST APIs, ONNX Runtime, Async Concurrency (Tokio), Multi-GPU Pipelined Inference
Data Science: NumPy, Pandas, Matplotlib
Tools: Git, GitHub, Jupyter Notebook, VS Code, LaTeX, Google colab

ACHIEVEMENTS

- Secured AIR 3426 among 1.5M+ candidates in JEE Main, achieving 99.9 percentile in Mathematics and ranking among the top 4% candidates in JEE Advanced.
- Ranked 8th of 224 teams (top 4%) in the Kaggle "LLM Classification Finetuning" competition, competing solo.

RELEVANT COURSEWORK

Data Structures & Algorithms (A)	—	Machine Learning (A-)	—	Scientific Computations (A-)	—	Deep Learning	
—	Probability & Statistics	—	Natural Language Understanding (NLU)	—	Mathematics-I	—	Mathematics-II